

Axisymmetric (RZ) Navier–Stokes in Cerisse: Mathematical Analysis of the Near-Axis Discretization and the Missing Viscous Geometric Terms

Cerisse SRP validation — internal technical note

June 17, 2026

Abstract

We derive the compressible Navier–Stokes equations in cylindrical (r, z) coordinates without swirl, cast them into the finite-volume “metric divergence + geometric source” form that Cerisse uses, and compare term-by-term against the implementation (`src/tim/compute_rhs.cpp`, `src/rhs/viscous.h`). Two distinct issues are identified and proven mathematically. **(i) Euler level:** no physical term is missing — the inviscid geometric source $+p/r$ is present and is *well balanced* for uniform pressure. However, the discrete radial-momentum balance is *not* well balanced *across a shock*: the cell-centre pressure used in the $+p/r$ source is inconsistent with the reconstructed face pressures inside the flux, leaving a spurious radial-momentum source whose amplitude is $[p_c - \frac{1}{2}(p_-^f + p_+^f)]/r_c$ and which is amplified by $1/r_c = 2/\Delta r$ in the on-axis cell. This seeds the spurious near-axis velocity at on-axis shocks and scales like $1/\Delta r$. **(ii) Navier–Stokes level:** the cylindrical viscous geometric terms are genuinely absent — the dilatation omits u_r/r , the hoop stress $\tau_{\theta\theta}$ is never formed, and the radial-momentum source $-\tau_{\theta\theta}/r$ is not added (`rz_geometric_source` is a no-op, and is additionally skipped for IBM/EB cases). This does not *seed* the artifact (it is present in inviscid runs) but removes the only physical sink that would damp it. The defect lies in the NS/RZ discretization, not in the IBM machinery.

1 Governing equations in cylindrical coordinates

Assume axisymmetry ($\partial/\partial\theta = 0$) and no swirl ($u_\theta = 0$). The velocity is $\mathbf{u} = u_r(r, z, t) \mathbf{e}_r + u_z(r, z, t) \mathbf{e}_z$. The compressible Navier–Stokes system reads

$$\frac{\partial \rho}{\partial t} + \frac{1}{r} \frac{\partial(r\rho u_r)}{\partial r} + \frac{\partial(\rho u_z)}{\partial z} = 0, \quad (1)$$

$$\frac{\partial(\rho u_r)}{\partial t} + \frac{1}{r} \frac{\partial(r\rho u_r^2)}{\partial r} + \frac{\partial(\rho u_r u_z)}{\partial z} + \frac{\partial p}{\partial r} = \underbrace{\frac{1}{r} \frac{\partial(r\tau_{rr})}{\partial r} + \frac{\partial\tau_{rz}}{\partial z} - \frac{\tau_{\theta\theta}}{r}}_{(\nabla \cdot \boldsymbol{\tau})_r}, \quad (2)$$

$$\frac{\partial(\rho u_z)}{\partial t} + \frac{1}{r} \frac{\partial(r\rho u_r u_z)}{\partial r} + \frac{\partial(\rho u_z^2 + p)}{\partial z} = \frac{1}{r} \frac{\partial(r\tau_{rz})}{\partial r} + \frac{\partial\tau_{zz}}{\partial z}, \quad (3)$$

$$\frac{\partial E}{\partial t} + \frac{1}{r} \frac{\partial(r u_r (E + p))}{\partial r} + \frac{\partial(u_z (E + p))}{\partial z} = \frac{1}{r} \frac{\partial(r(u_r \tau_{rr} + u_z \tau_{rz} - q_r))}{\partial r} + \frac{\partial(u_r \tau_{rz} + u_z \tau_{zz} - q_z)}{\partial z}. \quad (4)$$

The Newtonian stress tensor (Stokes hypothesis, bulk viscosity ξ) has the cylindrical components

$$\tau_{rr} = 2\mu \frac{\partial u_r}{\partial r} + \lambda\Theta, \quad \tau_{zz} = 2\mu \frac{\partial u_z}{\partial z} + \lambda\Theta, \quad \boxed{\tau_{\theta\theta} = 2\mu \frac{u_r}{r} + \lambda\Theta}, \quad \tau_{rz} = \mu \left(\frac{\partial u_r}{\partial z} + \frac{\partial u_z}{\partial r} \right), \quad (5)$$

with $\lambda = \xi - \frac{2}{3}\mu$ and the **cylindrical dilatation**

$$\Theta = \nabla \cdot \mathbf{u} = \frac{\partial u_r}{\partial r} + \frac{u_r}{r} + \frac{\partial u_z}{\partial z}. \quad (6)$$

Where the geometric source terms come from. Because $\mathbf{e}_r$ rotates with θ , the divergence operator carries the metric factor $1/r$, and two terms in the radial momentum equation (2) are *not* expressible as a flux divergence over the (r, z) faces:

1. the **inviscid pressure source** $+p/r$, obtained by writing the pressure inside the radial flux, $\frac{1}{r}\partial_r(rp) = \partial_r p + p/r$, so that

$$\frac{\partial p}{\partial r} = \frac{1}{r} \frac{\partial(rp)}{\partial r} - \frac{p}{r}, \quad (7)$$

2. the **viscous hoop source** $-\tau_{\theta\theta}/r$ in $(\nabla \cdot \boldsymbol{\tau})_r$.

Continuity (1), axial momentum (3) and energy (4) are *pure* flux divergences in cylindrical FV form (the $\mathbf{e}_z$ direction and $u_\theta \equiv 0$ produce no extra source). *All* geometric source terms therefore live in the radial momentum equation: one inviscid ($+p/r$) and one viscous ($-\tau_{\theta\theta}/r$).

2 Finite-volume formulation used by Cerisse

On a cell $[r_-, r_+] \times [z_-, z_+]$ with $r_\pm = r_0 + (i, i+1)\Delta r$, $r_c = \frac{1}{2}(r_- + r_+)$, the cylindrical control volume is $V = \pi(r_+^2 - r_-^2)\Delta z$ and the radial face area is $A_r = 2\pi r\Delta z$. The metric-consistent radial divergence is

$$-\frac{1}{r} \frac{\partial(rF_r)}{\partial r} \Big|_{\text{cell}} \approx \frac{2(r_-F_- - r_+F_+)}{r_+^2 - r_-^2} = \frac{r_-F_- - r_+F_+}{r_c \Delta r}, \quad (8)$$

which is exactly `compute_rhs.cpp:261`. Cerisse builds the face fluxes $F_\pm$ with the ordinary *Cartesian* momentum flux (WENO-Z5 characteristic Lax–Friedrichs); the RZ weighting is applied only in (8). The radial-momentum geometric source is then added as

$$\left(\frac{\partial(\rho u_r)}{\partial t}\right)_{\text{source}} = \frac{p_c}{r_c}, \quad \frac{1}{r_c} = \frac{2}{\Delta r} \text{ in the axis cell } (i=0), \quad (9)$$

(`compute_rhs.cpp:264--269`). The viscous geometric hook `rz_geometric_source` (`compute_rhs.cpp:311`) is called only when `not AMREX_USE_GPIBM/CNS_USE_EB`, and its body is empty (`src/rhs/viscous.h:587`, `viscousLES.h:628`).

3 Euler level: correct, but not well balanced across a shock

Write the discrete radial-momentum pressure contribution to the RHS, using the face pressures $p_\pm^f$ that the flux actually carries (from (8)–(9)):

$$\mathcal{R}_p = \frac{r_- p_-^f - r_+ p_+^f}{r_c \Delta r} + \frac{p_c}{r_c}. \quad (10)$$

The exact physical term is $-\partial p/\partial r \approx -(p_+^f - p_-^f)/\Delta r$.

(a) Uniform pressure $\Rightarrow$ exactly balanced. With $p_-^f = p_+^f = p_c = p$ and $r_- - r_+ = -\Delta r$,

$$\mathcal{R}_p = \frac{p(r_- - r_+)}{r_c \Delta r} + \frac{p}{r_c} = -\frac{p}{r_c} + \frac{p}{r_c} = 0 = -\frac{\partial p}{\partial r} \Big|_{\text{uniform}}. \quad (11)$$

The $+p/r$ source exactly cancels the spurious p/r generated by putting p in the radial flux (cf. (7)). Good.

(b) General pressure: the residual. Substituting $r_{\pm} = r_c \pm \frac{1}{2}\Delta r$ into (10) and adding the exact $-\partial p/\partial r$, the discrete *error* (the spurious radial-momentum source) is, after cancellation,

$$\boxed{\mathcal{E} \equiv \mathcal{R}_p - \left(-\frac{\partial p}{\partial r}\right) = \frac{p_c - \frac{1}{2}(p_-^f + p_+^f)}{r_c}} \quad (12)$$

Derivation. Form $\mathcal{R}_p + (p_+^f - p_-^f)/\Delta r$ and multiply by $r_c\Delta r$:

$$r_-p_-^f - r_+p_+^f + p_c\Delta r + r_c(p_+^f - p_-^f) = \frac{1}{2}(r_- - r_+)(p_-^f + p_+^f) + p_c\Delta r = \Delta r \left[p_c - \frac{1}{2}(p_-^f + p_+^f) \right],$$

using $r_-p_-^f - r_+p_+^f + \frac{1}{2}(r_- + r_+)(p_+^f - p_-^f) = \frac{1}{2}(r_- - r_+)(p_-^f + p_+^f)$ and $r_- - r_+ = -\Delta r$. Dividing by $r_c\Delta r$ gives (12). $\square$

Consequence.

- **Smooth flow:** a (near-)central reconstruction gives $p_c \approx \frac{1}{2}(p_-^f + p_+^f)$, so $\mathcal{E} = \mathcal{O}(\Delta r^2)$ — negligible.
- **At a shock:** WENO reconstruction is upwind-biased and limited, so $p_c \neq \frac{1}{2}(p_-^f + p_+^f)$ and $\mathcal{E} = \mathcal{O}(\Delta p_{\text{shock}})/r_c$. In the axis cell $r_c = \frac{1}{2}\Delta r$, hence

$$\mathcal{E}_{\text{axis}} \sim \frac{2[p_c - \overline{p^f}]}{\Delta r} \propto \frac{1}{\Delta r}. \quad (13)$$

This is the seed. A spurious radial-momentum source of size $\mathcal{O}(\Delta p)/\Delta r$ appears wherever a strong shock (the Mach disk, the bow shock) crosses the axis. It violates the symmetry condition $u_r(r=0) = 0$, generates a near-axis radial velocity and, through $\partial u_r/\partial z$, azimuthal vorticity. No physical PDE term is missing: (13) is a purely *discrete* well-balancedness defect of the singular coordinate, and it is *worse on finer grids*, $\propto 1/\Delta r$.

4 Navier–Stokes level: the genuinely missing physics

Cerisse evaluates the viscous stresses with the *Cartesian* dilatation (`viscous.h:256,472`)

$$\Theta_{\text{code}} = \frac{\partial u_r}{\partial r} + \frac{\partial u_z}{\partial z} = \Theta - \frac{u_r}{r}, \quad (14)$$

i.e. the cylindrical u_r/r term of (6) is **dropped**. Consequently every normal stress is in error by $\lambda u_r/r$:

$$\tau_{rr}^{\text{code}} = \tau_{rr} - \lambda \frac{u_r}{r}, \quad \tau_{zz}^{\text{code}} = \tau_{zz} - \lambda \frac{u_r}{r}, \quad \tau_{\theta\theta}^{\text{code}} = (\text{not formed}). \quad (15)$$

Moreover the hoop geometric source of (2) is absent:

$$\boxed{\left(\frac{\partial(\rho u_r)}{\partial t}\right)_{\text{viscous-geom}} + = -\frac{\tau_{\theta\theta}}{r} = -\frac{1}{r}\left(2\mu\frac{u_r}{r} + \lambda\Theta\right) \text{ is missing.}} \quad (16)$$

Both omitted contributions carry a factor $1/r$ (directly, or through u_r/r), so they are negligible in the bulk but become significant only in the immediate neighbourhood of the axis. The energy equation (4) needs *no* separate geometric source — the viscous-work vector $\boldsymbol{\tau} \cdot \mathbf{u}$ has no θ -component, so $\nabla \cdot (\boldsymbol{\tau} \cdot \mathbf{u})$ is captured by the metric divergence — but it inherits the wrong Θ_{code} in τ .

Role. Equation (16) is the *only* physical mechanism that diffuses near-axis azimuthal vorticity. Its absence does not create the artifact (the seed (13) is inviscid), but it removes the sink that would otherwise damp it.

5 Which defect causes the observed near-axis vortices?

The two defects are independent and their roles are distinct:

	Euler seed (13)	Missing viscous hoop (16)
Nature	discrete well-balancedness defect	missing physical term
Active in	Euler <i>and</i> NS runs	NS runs only
Effect	<i>creates</i> spurious u_r at on-axis shocks	<i>fails to damp</i> it
Scaling	amplitude $\propto 1/\Delta r$	—
Fix	well-balanced pressure (route A)	implement (6),(16)

Numerical evidence consistent with this analysis:

- Inviscid RZ jets (`no_diffusive_t`) already show a non-zero near-axis u_r at the on-axis Mach disk $\Rightarrow$ the seed is *inviscid*, ruling out (16) as the cause.
- A Cartesian $M=6$ cylinder with the *same* WENO-LF flux has a smooth, clean bow shock (HLLC, by contrast, carbuncles) $\Rightarrow$ the flux family is not the culprit; the distinguishing factor is the RZ axis.
- The artifact is stronger at finer resolution (prominent at $L4$, weak at $L2$), matching the $1/\Delta r$ scaling of (13).

The bow-shock / Mach-disk *stand-off distances* are unaffected, because (13) perturbs the on-axis momentum balance, not the global shock position. The defect is in the NS/Euler RZ discretization; the IBM ghost-point machinery acts in the solid-adjacent cells, away from the open subsonic pocket where the vortices sit, and is not the driver.

6 Implied corrections

Route A — well-balanced pressure (removes the seed).

Discretize the radial pressure term so that (12) vanishes by construction, i.e. remove p from the radial-momentum flux and add a *cell-centred, consistent* gradient,

$$\left(\frac{\partial(\rho u_r)}{\partial t}\right)_{\text{pressure}} = -\frac{p_+^f - p_-^f}{\Delta r}, \quad p_\pm^f \text{ the same face pressures used in the flux,} \quad (17)$$

which is well balanced for arbitrary p and carries no $1/r$ amplification.

Route C — complete the RZ viscous operator (restores physics).

Use the cylindrical dilatation (6) in all stresses, form $\tau_{\theta\theta}$, add the hoop source (16) to the radial momentum, and remove the `AMREX_USE_GPIBM/CNS_USE_EB` gate at `compute_rhs.cpp:311`. This provides the physical damping of near-axis vorticity and makes RZ runs true Navier–Stokes.

Summary. At the Euler level nothing physical is missing; the spurious near-axis velocity is a discrete *well-balancedness* failure of the $+p/r$ geometric source across shocks, eq. (12), amplified by $2/\Delta r$ on the axis. At the Navier–Stokes level the cylindrical viscous terms (u_r/r in Θ , $\tau_{\theta\theta}$, and $-\tau_{\theta\theta}/r$) are genuinely absent, eq. (16), removing the physical sink for the seeded perturbation. Both are properties of the NS/RZ discretization, not of the immersed boundary.